

Towards Enhanced Breast Cancer Prediction: A Comparative Evaluation Study

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Abstract—Breast cancer remains a foremost cause of mortality among and prevalent malignancy globally, constituting 30% of all cancer diagnoses in women as of 2022. Timely detection of cancer before metastasis and accurate diagnosis of tumors as malignant poses a significant challenge in healthcare. Deploying accurate predictive models is crucial for facilitating life-saving early interventions. This study presents a comparative evaluation of Machine Learning (ML) classification algorithms for tumor type through the analysis of patient tumor structure quantification. Our research employed supervised ML algorithms that distinguish between malignant and benign tumors using tuning parameters to discern the optimal configurations for multiplying precision and minimizing the error rates. Additionally, a comparative analysis of K-Nearest Neighbors (KNN), Logistic Regression (LR), and Random Forest (RF) algorithms is carried out. The study's findings highlight the superiority of the RF Classification model, which attained an impressive 97% accuracy, surpassing KNN and LR, with recorded accuracies of 96% and 94%, respectively. Furthermore, performance metrics such as accuracy, precision, recall, f1-score, true and false positive and negative rates, and area under the Receiver Operating Curve (ROC) were applied for a comprehensive analysis of the algorithms. The results reveal improved accuracy compared to existing literature by 0.9%. Although limited to a single dataset, our methodology shows promising results for real-world applications using analogous input parameters. Future research should aim for the application of advanced tuning encompassing Deep Learning and Neural Networks.

Index Terms—Breast cancer detection, machine learning, classification, K-Nearest Neighbors, Logistic Regression, Random Forest.

I. INTRODUCTION

Engaging advanced statistical methods and computational algorithms, predictive modeling employs historical and current information with intent, to envision future events. This approach transpires a transformative shift in healthcare, introducing an instrument of diagnostics, optimizing patient outcomes by enabling cognizance of disease progression, improving response times for treatment (An et al., 2023), and aiding health outcomes with unprecedented precision. The significance of predictive modeling in healthcare is further highlighted by its exponential growth over the past decade. The integration of big data analytics and computational algorithms has opened new avenues for disease prediction and management, particularly in chronic and life-threatening conditions like cancer. Through utilizing substantial medical records, prognostic models bid a distinct comprehension of cancer histology, liable causes (Hassan et al., 2023), as well as probable ramifications of curative mediation, creating an abundance of customized and aggressive inferential models. Mammary tumors, a notorious plague touching mothers and daughters alike globally encircles convoluted characteristics, comprising of an assortment of strains inside the breast parenchyma. The female chest is made up of parenchyma variations, including a spectrum consisting of body fat to tightly arranged tissue, assorted into lobules

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which are milk-producing glands akin to the nipple by ducts. A complex organization made up of lymphatic systems, blood vessels, as well as connective tissue, which adheres to primary functions within tissue nourishment also waste removal. Mammary tumors can be categorically distinct between malignant and benign, through groundbreaking anatomical discoveries leading to curative treatment. Mammary tumors that are malignant command the capability of multiplying within other areas of the body, these tumors can be referred to as invasive lobular and ductal carcinoma. Invasive breast cancers, which include subtypes like invasive ductal carcinoma and invasive lobular carcinoma, are known for their ability to spread beyond the primary tumor site. Non-invasive or in situ breast cancers, such as ductal carcinoma in situ (DCIS) and lobular carcinoma in situ (LCIS), remain confined to their original location without invading neighboring tissues. This detailed understanding of breast cancer underscores the significance of early detection and precise classification, vital for optimizing treatment strategies and improving patient outcomes. The diversity in breast cancer types necessitates tailored diagnostic and therapeutic approaches (Ghorbian Ghorbian, 2023), highlighting the importance of ongoing research and technological advancements in this field. At the heart of mortality rates and increasing tumor sizes, stand younger women, where for them, time is not on their side as tumors grow exceedingly fast within, certainly double as to older women. The vitality of time is understated within the relationship of invasive tumors and their interactions with the host. Making the consequence of diagnostic delays soft-pedaled. Medical blunders are attributed highly to unattended patients and missed diagnostics, as a result of lack of suspicion towards malignancy. Leading to despairing treatment urgencies and uprising cancer stages toward inevitable mortality (Costa et al., 2024). These frequent yet statistically reported events lead to mistrust within the medical system and more importantly, highlight the necessity of a prediction model with high yet close to perfect accuracy. In the advancement of predictive modeling within healthcare, the utilization of ML techniques such as RF, KNN, and LR plays a pivotal role. RF's strength lies in its ensemble approach, utilizing multiple decision trees to enhance accuracy and prevent overfitting. LR, widely applied in risk prediction, offers a robust statistical analysis framework. Meanwhile, KNN's simplicity, based on the proximity of data points, proves effective in straightforward scenarios. These ML techniques, by harnessing diverse computational strategies, contribute significantly to the refinement of predictive models, aiming for heightened precision in disease diagnosis and prognosis, thereby underlining the transition towards more targeted and effective healthcare interventions (Huang et al., 2024). The use of these algorithms adds an effective tool for statistical analysis and risk predictions of data, very similar in approach, as compared to the layout of tumor quantification parameters of patients used within this study. The objective of this study is to incorporate ML techniques to aid medical and healthcare diagnosis (Gouda, 2022), which require urgent solutions, especially within rural and less advanced societies

where cost factors and qualified medical personnel remain a rare commodity as well as to elaborate on the effectiveness of ML through the enhancement of data-science to achieve models with appropriate industrial and medical application (Costa et al., 2024) and to allow widespread of diagnostic availability through the use of automation. The remainder of this paper is organized as follows with section II withholding the literature review, unveiling details of the studies on the algorithms applied within this paper. Section III stores the proposed methodology, which describes the phases of development, starting from data-pre-processing, feature scaling, feature selection, model selection, model parameter tuning to model evaluation. Section IV, implementation, goes into the integration and explanations of the various ML algorithms used within the paper. Section V, results and discussion, evaluates the performances of the models, using various evaluation metrics, and finally, Section VI discusses future works as well as limitations.

II. LITERATURE REVIEW

In this section, predictive machine learning algorithms used in the existing literature are discussed to unveil the workings and advantages behind the chosen algorithms, as well as the sources where each algorithm was used, highlighting their contribution to the study. Identify applicable funding agency here. If none, delete this text box. Gupta Garg (2020) have utilized K-Nearest Neighbour (KNN) algorithm which plots data rows on a vector plane based on the values of each feature respectively. Clusters are then created by grouping plots based on the Euclidean distance from each cluster. The number of K clusters can be selected arbitrarily but data visualizations can be generated to obtain an optimal amount of cluster that returns the best performance. As such, an advantage of KNN is that it requires minimal to zero model training before implementation, however, K values and feature selection must be set precisely to ensure its reliability when performing such an important classification. An implementation of classification on the WBCD with this algorithm obtained 95.8% accuracy with a K value of 6. Another common algorithm in the existing literature is Logistic Regression, which is a statistical algorithm that plots the parameter values to obtain its gradient with a cut of value. In this way, a cut-off is implemented to binarize the classification output (Houssami et al., 2019). This is suitable for breast cancer detection as the desired output is the 'malignant' and 'benign' label from the inserted data (Nanglia, 2022). Multi-parameter or multidimensional datasets will be calculated based on the coefficient value of each parameter hence model training is required to fine-tune the respective coefficients which represent the likelihood of affecting the classification based on its recorded value. This model achieved an implementation with a maximum accuracy of 95.8% (Gupta Garg, 2020). Among decision trees algorithms, Random Forest classifier has been applied in the literature. It generates a logical tree structure based on parameters for feature classification, Random Forest Classifiers generate several decision trees based on the same parameters. It then

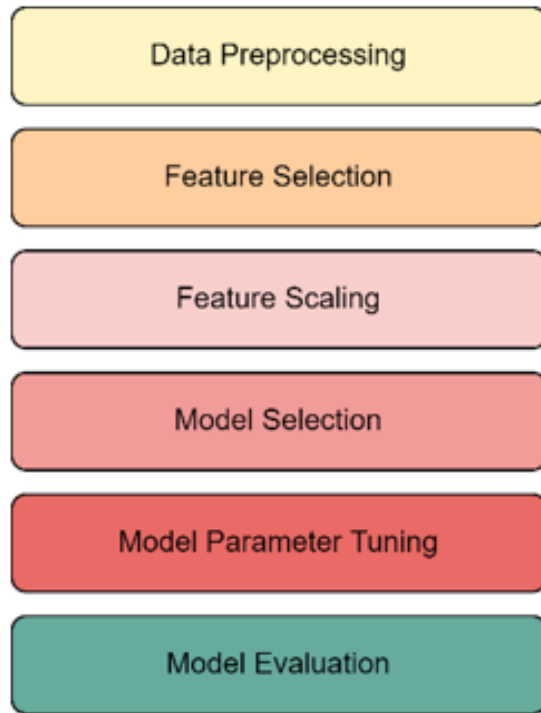


Fig. 1. Flowchart of Proposed Methodology

returns the resulting classification from the input by selecting the most likely value based on the statistics of results returned by the generated trees (Octaviani Rustam, 2019). Hence, this algorithm should in theory have a more robust performance as compared to decision tree classifications. Furthermore, there is no need for pruning as obtaining results from multiple trees should statistically reduce overfitting tendencies and errors from individual trees. An implementation based on this algorithm yielded 100% accuracy when 90% of the dataset was allocated for training (Octaviani Rustam, 2019). However, Decision Tree Classifiers have also been successfully applied for this use case such as the implementation of Sindhu et al., (2017) albeit at a lower accuracy of 75.87%.

III. PROPOSED METHODOLOGY

The proposed methodology summarizes the steps and methods taken to build the supervised machine learning models to achieve the results shown within the results and discussion in section IV.

Fig. 1. . Several formats of open-source datasets are available for model training and testing depending on the type of classification model generated. These forms include images of breast mammograms and measurement values of breast mass in structured information format (XML). Common image datasets include the CBIS-DDSM mammogram dataset (Hekal et al., 2020), while measurement datasets include the Breast Cancer Wisconsin (Diagnostic) available from the UCI Machine Learning Repository (Wolberg, 1995). This study

utilized the Diagnostic Wisconsin Breast Cancer Database, due to its wide range of features available such as cell texture and concavity.

IV. PHASE 1 - DATA PREPROCESSING

In the initial phase, pre-processing takes place. To optimize predictive analytics, a range of pre-processing techniques are employed to enhance data quality. This involves scaling feature values to optimize algorithms within the model. The primary procedural steps consist of data transformation, cleaning, and reduction. The importance of partitioning datasets into dependent and independent variables before pre-processing to prevent data leakage is understated. Typically, missing values are addressed by either removal or substitution with the mean or median of the feature. In this context, replacing missing data with the mean is deemed more suitable, as it preserves the limited number of rows in the dataset, enabling more data to be processed by the algorithm. On the contrary, noisy data often manifest as outliers, exhibiting extremely high or low values that can disrupt the feature's mean and detrimentally impact the algorithm's ability to discern meaningful patterns and trends. As a result, the importance within the pre-processing phase of eliminating outliers is absolute. They often stem from cognisant errors in data collection and can include missing values. Interquartile range or IQR stands as one of the many options of implementations in pre-processing for sifting out outliers within data. Among others are such as standard deviation or percentiles. In choosing an implementation, several factors tied to the characteristics of the pool of data, come into consideration. Standard deviation is an appropriate option for data with normal distribution, otherwise, outlier removal through the means of interquartile ranges or percentiles would be preferred, leading towards data of skewed distributions. Often applied to the target class or dependant variable, label encoding, is a technique in which categorical values in writing are converted to numerical ones, where they can be processed and compared easier. As machine learning algorithms function through the processing of numbers this method is evidently beneficial. Although using such a technique shines more within situations of multinomial classification, yielding an increased number of target classes within its data. In order to carry out anomaly removal of data, a function is defined to perform outlier removal using IQR, this is done by sifting the difference between the third and first quartile within a sample of data. To define the Lower boundaries, one and a half times the IQR is subtracted from the first quartile value, whereas the upper boundaries are defined by adding one and a half times the IQR toward the third quartile value. Thus, the remaining samples which lay beyond the boundaries are sifted out.

A. Phase 2 – Feature Selection

Within phase 2, to produce data of high quality in fielding a robust prognostic model, feature selection is carried out as a standard of predictive modelling. Data reduction stands as one of the many steps to be taken in working towards this priority. To do so, features containing a lower correlation

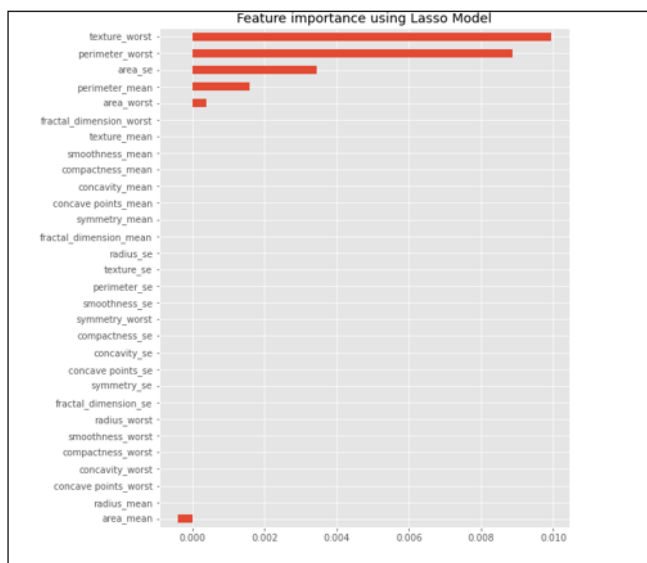


Fig. 2. Feature Selection for Correlated Variable

to the dependent variable of the model are removed. It is also beneficial to retain features that correlate negatively, as they add to the discriminative ability of the prognostic model. LASSO regression is selected as the preferred feature selection approach. This is because the methods sifts out many of the models' overall features upon usage, this allows just a small number of features to remain, which happen to be the ones that are highly correlated to the dependent variable. Such an aspect proves reliable upon datasets, which consist of a large amount of features. The LASSO regression methods ensures that the features used for training the model result in optimized results due to the quality of data in correlation to their correlation to the dependent variable. The illustration in Figure 2, displays the usage of an alpha metric used to organize the coefficients of features within the model representing their relevance to the target class. The alpha works as an error measure, comparing the variance of residuals within a best-fitted line, thereby, adjusting the coefficients of features to the residual sum of squares. The alpha also plays a part in ensuring the magnitude of coefficients is balanced. The coefficients given to the individual features, tell us the importance it will have in aiding the inference of the prognostic model. Insignificant values are discarded whereas highly relevant ones are kept signalling their importance.

The importance of the LASSO regression lies in its capability to autonomously remove unnecessary features from the model, rendering the need for a manual selection to be redundant. In its calculations, the LASSO regression removed eighty percent of features deeming them unnecessary in consideration of the model's benefit. By segregating individual data frames for the target class and features, the LASSO regression may be fitted.

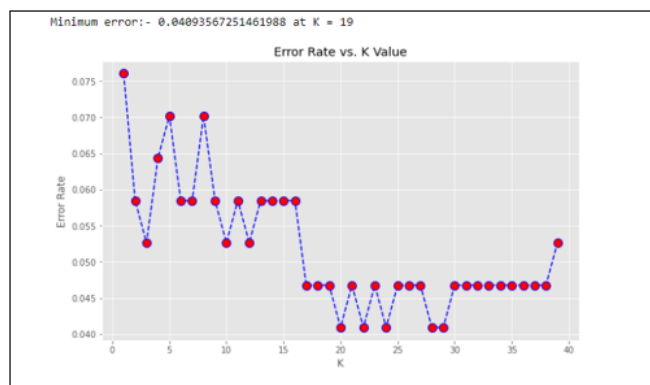


Fig. 3. Illustration of Error Rate for each K value

B. Phase 3 – Feature Scaling

Feature scaling is used with the intent to further reduce noise within the pool of selected data. In order to optimize the prognostic model's accuracy, minMax scaling is adopted for the prerogative by standardizing all sample values toward a range of 0 and 1. This approach works better for some algorithms compared to others. Standardizing and scaling the values of data, work particularly well for the selection of algorithms selected within this study. It facilitates the concept of distance amongst the neighbors of the K-nearest neighbors and adheres soundly towards the probability distribution of logistic regression which is binomial. It also inherently works well with the decision tree thresholding prospected within the classification of Random Forest. Another benefit to fielding this technique is that visualizations appear very well as their values share a boundary of range.

C. Phase 4 – Model Selection

The use of the selected algorithms fit the criteria for the prognostic model, examined with this study. The focus if the selection illustrates a plethora of supervised machine learning models, which compared well against one another due to the binomial scaling and nature of their calculations, in relation to the configurations of the dataset. The The selected binomial classification algorithms for the predictive analytics framework include the Random Forest Classifier, Logistic Regression, and K-Nearest Neighbours.

D. Phase 5 – Model Parameter Tuning

Within the model parameter tuning phase, the parameters of algorithms are tested to discover the best configurations for optimizing model accuracy. This can be done by running iterations of training and comparing the error rates of their results using different values of parameters for various algorithms. As illustrated in Figure 3, the value of K at 19 yields a minimal error rate observed within the iterations of numerous attempts configured with different values of parameters. Yielding a result of 0.04, the parameter tuning concludes that using those settings, only four incorrect predictions are captured out of 100.

This analytical approach provides a comprehensive view of the model's performance across various K values, aiding in the selection of an appropriate parameter for optimal accuracy. A similar method is also deployed to tune the number of estimators best fitted for the RF classifier within the context of the pre-processed data to be inferred. The range of estimator values tested is between 64 and 128 as they provide a good organization between processing times also in regards to the area under the receiver operating curve. The tuning of the RF classifier revealed a peak outcome of a 3% error rate.

E. Phase 6 – Model Evaluation

In order to review the efforts within this study, the implementation is tested through various rigorous metrics that aim to illustrate the performances of the model in sufficient areas. This includes how the selection of classifiers and the approach of pre-processing have affected the overall performances of the prognostic outcomes through different test criteria. These include methods such as a fundamental of confusion matrices which tell us the number of outcomes correctly anticipated for positive and negative expectations of the model. From here the received values help to elevate the calculations to metrics such as precision, recall, and accuracy, which can be analyzed to form hypotheses upon the various choices made within the building of the prognostic model.

V. IMPLEMENTATION

The implementation segment of the study details the methodology of training the machine learning algorithms with the preprocessed data, in order to output a prognostic model, which can be compared among the selection of machine learning algorithms in question. The independent and dependent variables are entered within the train-test split utilizing the sklearn library. The incorporation of a random state is imperative to maintain consistency in the values of randomization for sampling. This consistency enables the reliable use of various data subsets from both the training and testing sets for making predictions and generating model visualizations. The systematic application of a random state ensures stability in the evaluation and interpretation of the model's performance across different runs. Algorithm:

- Fit the dataset and divide it into training and testing sets.
- Tune perimeters of the algorithm for optimal accuracy
- Visualize and evaluate the results of model selection.

A. Logistic Regression

Logistic regression, in contrast to linear regression, follows a similar approach outputting a best-fitted regression line. Subsequently, the established generations of dependent variables of features are passed through a sigmoid function whereby an outcome is decided. Using a predetermined threshold the outcome is sifted towards binary finalizations, and based on this comparison, values are classified as either 1 or 0. The application of logistic regression can be visualized through a sigmoid function, particularly useful in two-dimensional

scenarios" (says, 2020). General workflow for Logistic Regression:

- Fit the dataset and divide it into training the g and testing sets.
- Visualize and evaluate the results of model selection.

VI. K-NEAREST NEIGHBORS

K-Nearest Neighbors calculates the distance between data points and their nearest neighbors, using Euclidean distance. This is done through a parameter termed 'k' which decides the number of data points considered, within its calculations. It uses a majority count to determine the class of a data point, compared to the amount of nearest neighbors by k. KNN performs better within undetermined data distribution whereas classifiers such as LR only perform well under Gaussian distributions. General workflow of K-nearest Neighbor:

- Tune the optimal k-value for the dataset.
- Input training data and optimal parameters into the algorithm
- Evaluate and print the accuracy of the model performance.

A. Random Forest

Random Forest is a supervised machine learning algorithm comprised of regression trees and an ensemble of unpruned classification for making predictions through aggregation. The Random Forest classifier employs multiple decision trees, often referred to as estimators, to construct a robust prediction model. Each estimator individually classifies samples based on distinct independent variables. The final prediction is determined by aggregating the outcomes of the majority of decision trees, resulting in a more reliable and accurate classification. This ensemble approach enhances the overall predictive power and resilience of the Random Forest algorithm" (Kumar, 2020). General workflow of Random Forest:

- Tune the optimal estimator value for the dataset.
- Input training data and optimal the parameters into the algorithm
- Evaluate and print the accuracy of the model performance.

VII. RESULTS DISCUSSION

The following section encompasses the model evaluation for assessing the performance of the chosen machine learning algorithms: Random Forest, Logistic Regression, and K-Nearest Neighbors in the prediction of breast cancer detection. Additionally, this section covers the conclusion, limitations, and potential avenues for further research within the paper.

As illustrated in Figure 4, the evaluation of the three algorithms employed in the model reveals distinct accuracies. The Random Forest classifier demonstrates superior performance among the three models, achieving an accuracy of 97.08%. Following closely, K-Nearest Neighbors secures the second-highest accuracy at 95.91%, while Logistic Regression trails with an accuracy score of 94.15%. In Figure 5, the ROC curve representations for each algorithm are presented in descending

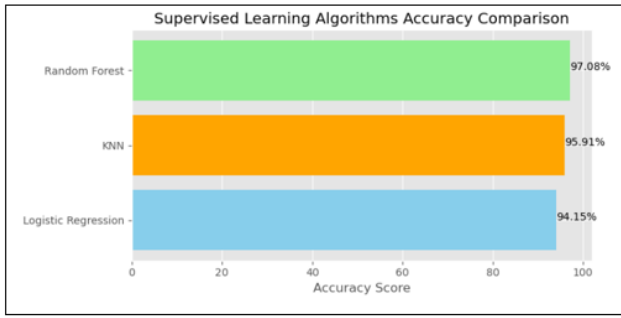


Fig. 4. Accuracy result comparison of supervised ML algorithms

order based on their AUC values. The Random Forest curve performs well with an impressive AUC of 0.95, representing its strong discriminatory ability. Following closely is Logistic Regression with an AUC of 0.94, while K-Nearest Neighbors lags slightly behind with an AUC of 0.91.

These AUC values align with the overall performance of each algorithm in the breast cancer prediction model. The Random Forest Classifier, which boasts the highest accuracy at 97.08%, also exhibits the most robust discrimination ability as evidenced by its leading AUC. The KNN classifier retained the lowest AUC and boasts an accuracy still higher than the LR classifier. This entails that the LR classifier holds a better ability to discriminate classes as compared to the KNN classifier although its accuracy is lower. This phenomenon occurs as the LR classifier yielded a better false positive rate compared to the KNN classifier within the prognostic outcome. The AUC offers significant insights into measuring the ability of a classifier to distinguish between classes, allowing a selection between accuracy and discriminatory capabilities for a selected application. In evaluating the output values of the confusion matrix for the various algorithms, its precise that both the RF as well as LR classifiers performed exceedingly well in producing correct outputs for positive outcomes, both generating 109 outcomes with only one wrong output. However the LR classifier performed poorly in generating results for negatively expected outcomes, whereas the RF classifier remained consistent. In predicting results for negatively expected outcomes, the KNN classifier and RF both performed in generating 57 outcomes correctly whilst wrongly generating four outcomes.

With the attained outcomes of the confusion matrix, a classification report can be established as illustrated in Figure 6. The classification report furnishes comprehensive metrics, such as f1-score, precision, and recall, for each algorithm, thereby aiding in accuracy computation. The factored mean between precision and recall metrics can be considered as an f1-score. Precision signifies the ratio of correctly predicted positive values to falsely predicted positive values, while recall represents the ratio of correctly predicted values over the total number of actual positive outcomes. As can be visualized in Figure 7 according to the findings of Haq et al. (2023), it

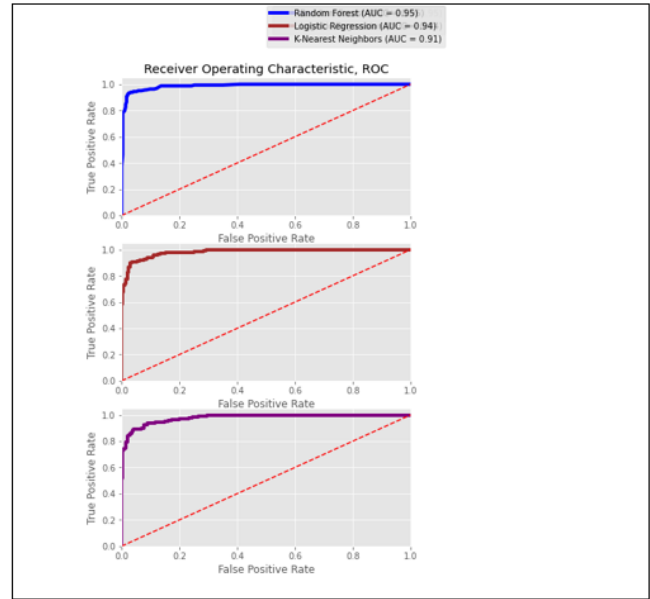


Fig. 5. Algorithms ROC Curve

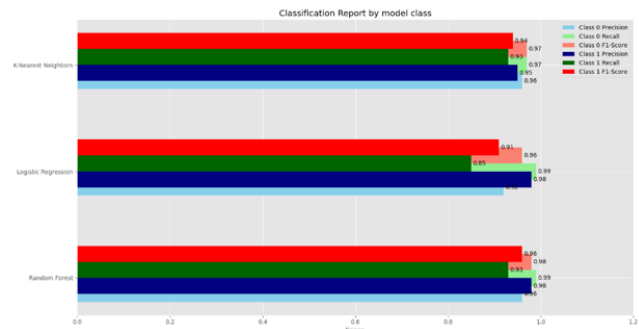


Fig. 6. Visualisation of the Printed Classification Reports

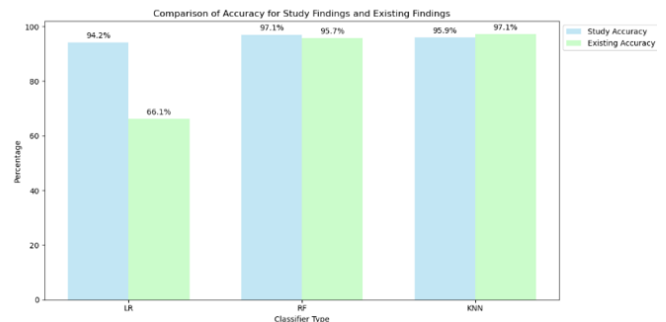


Fig. 7. Study Findings vs existing findings.

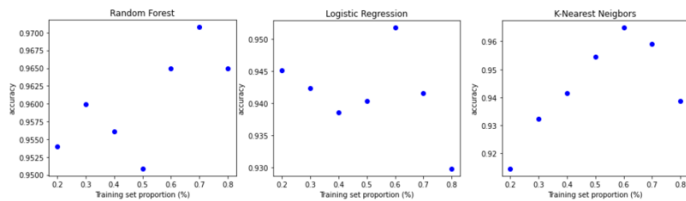


Fig. 8. Training Set Proportion of the Algorithms.

can be observed that the accuracies of the models achieved within this study outperform the findings of existing research within the space, which can be attributed to the execution of data science and hyperparameter tuning, of the models to, increase and optimize model accuracies and recalls. The noticeable improvements are large for LR with a 28.1% improvement, and less noticeable however, still well improved for RF, which can be argued to be even more impressive, as the RF model, performs very well in many studies, (above 90%), improving it still, proves to be challenging and more effective improvements, as with already high accuracies, the results, upon an improvement rate of 1.4%, can be implemented for employable scenarios within the diagnosis, proving that the following implementation of data science and parameter tuning was superior to existing findings.

VIII. CONCLUSION

In conclusion, the Random Forest model performed the best for the selected application among the three algorithms tested, from all evaluation metrics, the random forest algorithm, scored an accuracy of 97% with a true positive value of 109 and a false positive value of 1 as well as a true negative value of 57 and a false negative value of 4, which scores the least errors between the three algorithms. As well as having the best sensitivity at 99% and specificity at 93%. It can also be noted that through more advanced perimeter tuning which includes a wider spectrum of perimeters, the random forest algorithm may still be improved in performance for the selected dataset and application. A limitation noticed in the works of the model is that the training set proportions about the accuracies of the individual models were evaluated after building the models, this created inefficiencies within the working model in how well they could perform, as can be seen in Figure 8, accuracies of the model could be further enhanced. Another limitation that stands is, that the parameter tuning is not optimal, using techniques such as GridSearchCV, the algorithms may be fully enhanced through thorough and full-scale parameter tuning. In the future, the plan is to incorporate the training set proportion evaluation as a part of hyperparameter tuning before the prognostic inference, thereby, maximizing its reliability. Furthermore, Convolutional Neural Networks (CNNs) with transfer learning can be used for computer vision-based classification and enhanced tuning.

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